

CS & IT ENGINEERING



Computer Network

Routing

Lecture No. - 06



By - Abhishek Sir



Recap of Previous Lecture



Topic

Distance Vector Routing

Topic

Routing Protocols



Topics to be Covered



Topic

Hierarchical Routing

Topic

Source Routing

Topic

Traffic Shaping

ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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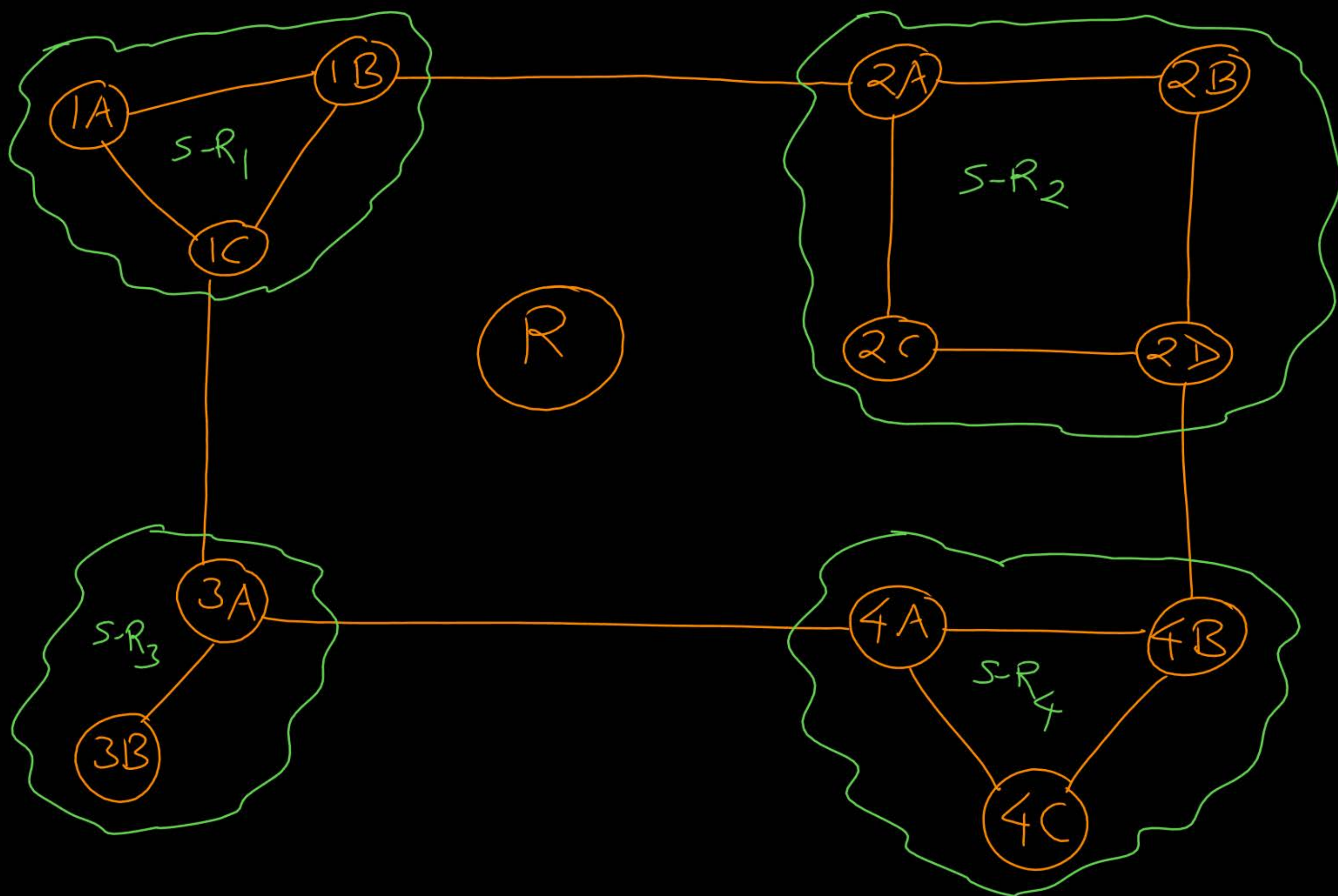




Topic : Hierarchical Routing



- The network is divided into regions and sub-regions.
[To create hierarchy]
- Allow efficient route management
- Route Aggregation ✓
[To reduce the size of routing tables]



Router 1A, global routing Table



Destination	Next Hop
1A	1A
1B	1B
1C	1C
2A	1B
2B	1B
2C	1B
2D	1B
3A	1C
3B	1C
4A	1B/1C
4B	1B/1C
4C	1B/1C

Router 1A, Hierarchical Routing Table



Destination	Next Hop
1A	1A
1B	1B
1C	1C
2	1B
3	1C
4	1B/1C

[MSQ]

IIT-K [GATE-2023][1 Mark]



#Q. Which of the following statements is/are INCORRECT about the OSPF (Open Shortest Path First) routing protocol used in the Internet?

- ✓ A OSPF implements Bellman-Ford algorithm to find shortest paths. FALSE
- B OSPF uses Dijkstra's shortest path algorithm to implement least-cost path routing. TRUE (OSPF-LS)
- ✓ C OSPF is used as an inter-domain routing protocol. FALSE (Intra-domain)
- D OSPF implements hierarchical routing. TRUE

Ans: A & C



Topic : Source Routing



- Path addressing
- Source host determine the route (partial or complete) for a packet
[before transmission]
- Each packet carry determined routing path in optional IP Header

#Q. Consider the following statements about the functionality of an IP based router.

- I. A router does not modify the IP packets during forwarding. FALSE
- II. It is not necessary for a router to implement any routing protocol. TRUE
- III. A router should reassemble IP fragments if the MTU of the outgoing link is larger than the size of the incoming IP packet. FALSE

Which of the above statements is/are TRUE ?

[GATE-2020, 1-Mark]

(A) I and II only

(B) I only

(C) II and III only

✓ (D) II only

Ans: D

IIT-D



Topic : Traffic Shaping



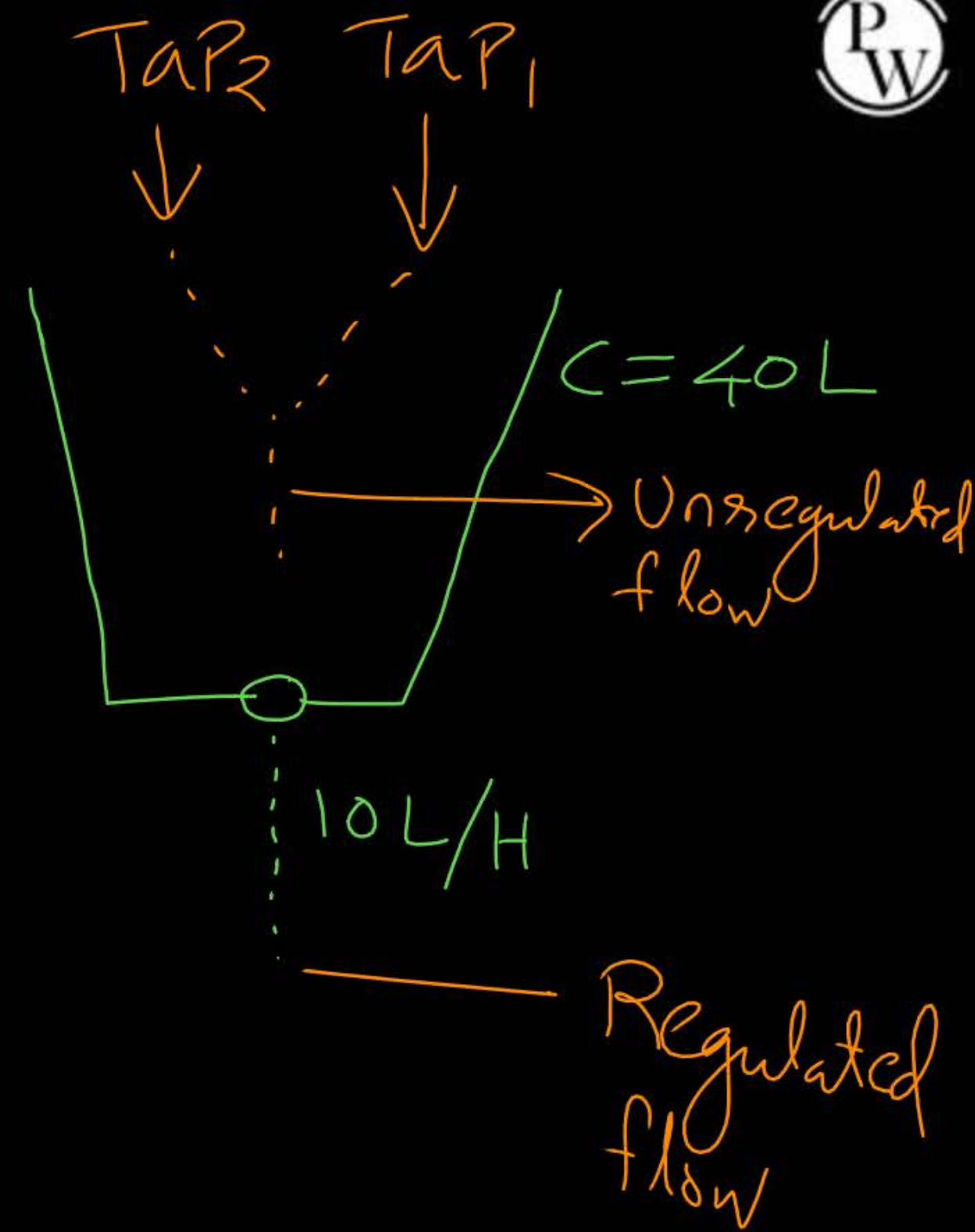
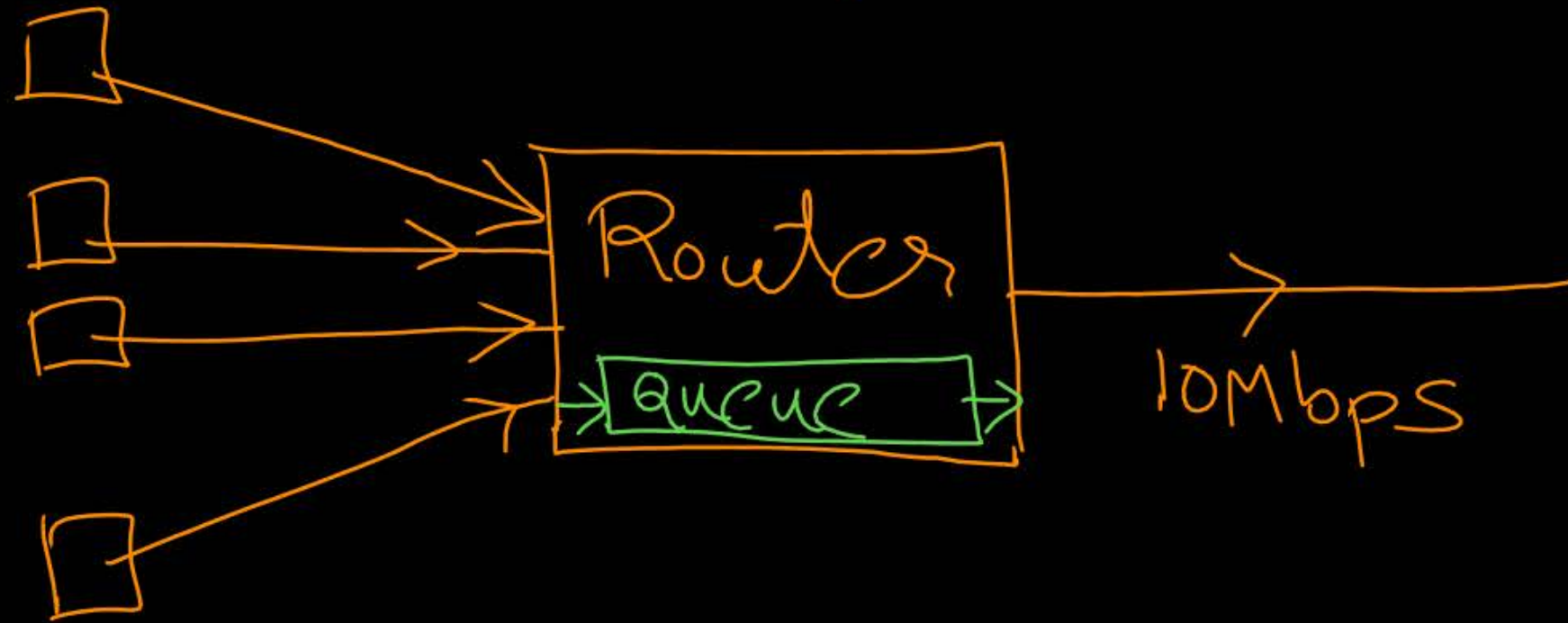
- Congestion control at router 
[Congestion occur due to bursty traffic]
- Two traffic shaping algorithm :-
 1. Leaky Bucket Algorithm
 2. Token Bucket Algorithm



Topic : Leaky Bucket Algorithm



- Convert unregulated flow into regulated flow
- Outflow is at a constant rate





Topic : Token Bucket Algorithm

→ Sometimes needed to speedup the output rate for a while

C = Capacity of token bucket (in bytes)

R = Token arrival rate (in bytes per sec)

M = Maximum output rate (in bytes per sec)

S = Duration for maximum output rate permissible (in sec)

$$(R < M)$$

$$[C + R * S] = M * S$$

$$S = \frac{C}{(M - R)}$$

#Q. A computer on a 10Mbps network is regulated by a token bucket. The token bucket is filled at a rate of 2Mbps. It is initially filled to capacity with 16Megabits. What is the maximum duration for which the computer can transmit at the full 10Mbps?

[GATE 2008]

115C

(A) 1.6 seconds

✓ (B) 2 seconds

(C) 5 seconds

(D) 8 seconds

Ans: B

Solution :-

$$C = 16 \text{ Mb (Megabits) (Full)}$$

$$M = 10 \text{ Mbps}$$

$$R = 2 \text{ Mbps}$$

$S =$ Maximum duration for which
the computer can transmit at the full capacity

$$S = \frac{C}{(M-R)} = \frac{16 \text{ Mb}}{(10 \text{ Mbps} - 2 \text{ Mbps})} = \frac{16 \text{ Mb}}{8 \text{ Mbps}} = 2 \text{ sec}$$

#Q. For a host machine that uses the token bucket algorithm for congestion control, the token bucket has a capacity of 1 megabyte and the maximum output rate is 20 megabytes per second. Tokens arrive at a rate to sustain output at a rate of 10 megabytes per second. The token bucket is currently full and the machine needs to send 12 megabytes of data. The minimum time required to transmit the data is _____ seconds.

if token Bucket is currently Empty

[GATE 2016]

$$R = 10 \text{ MBps}$$

$$\boxed{\text{Data Size} = 12 \text{ MB}} \quad 11 \text{ SC}$$

$$10 \text{ MB} \rightarrow 1 \text{ sec}$$

$$12 \text{ MB} \rightarrow \frac{12}{10} * 1 \text{ sec}$$

$$\rightarrow 1.2 \text{ sec}$$

Solution :-

$$\text{Data Size} = 12 \text{ MB}$$

$$C = 1 \text{ MB (Megabytes)} \text{ (Full)}$$

$$M = 20 \text{ MBps}$$

$$R = 10 \text{ MBps}$$

S = Maximum duration for which
the computer can transmit at the full capacity

$$S = \frac{C}{(M-R)} = \frac{1 \text{ MB}}{(20 \text{ MBps} - 10 \text{ MBps})} = \frac{1 \text{ MB}}{10 \text{ MBps}} = \frac{1}{10} \text{ sec} = 0.1 \text{ sec}$$

$$\text{Minimum time required to transmit the data} = S + \frac{(\text{Data Size} - M \times S)}{R}$$

$$= 0.1 \text{ sec} + \frac{(12 \text{ MB} - 20 \text{ MBps} \times 0.1 \text{ sec})}{10 \text{ MBps}} = \boxed{1.1 \text{ sec}}$$



2 mins Summary



Topic

Hierarchical Routing ✓

Topic

Source Routing ✓

Topic

Traffic Shaping ✓



THANK - YOU

